Meeting with John Wood White Plains Nov 18, '58

IDP.

- no big studies yet.

8 factors which will effect computing requirements
(7 of them say larger & faster, 1 doesn't.)

IDP wants to use any machine IBM builds — may need 2, one for standby.

1. addition capacity to hold supervisory program to solve multiprograming problem.
2. Often a single input transaction will be followed thru all the master records to which it applies. Machine will need programs for all the jobs on hand.
3. Need more redundant equipment to guard against breakdown (additional computer or extra checking in one computer.)
   - real time requirement —
4. When one has real time capability many customers will want to make more frequent cycles of same old applications eg. forecast daily not weekly — more current data.
5. Existance of more current data will bring about new applications & — more detail or scientific management reports
6. Disks vs tapes: On an IDP with disks - random access has less compute time between data accessing ops — access is faster so computering must be faster too.

7. Real Time - sufficient comp. capacity to handle peak rates
just average rates.
(will be in STAFF Report Early Dec will be out)
one to decrease regs.

8. In todays - batch processing there are peak loads eg. payroll
at end of week — IDP will tend to smooth out these peaks.

1: Variety of IDP applications.

made survey of 23 650-Remote — none close to IDP

- just replacing disk files for tapes.
- 305 applications are not either
- 310 probably not unless better data collection system is provided

---

IDP system will have 3 main sections:

1. Computer
2. I/O
3. Storage - Fast-Random access (faster than tapes)
100 ms

The above 3 should be coupled as loosely as possible. For particular customer - different configuration — can use any computer.
However File will require separate computer for its control (may be 310?)
+ may need computer tied to I/O also (STRETCH xchange?)
(File Control: 5000 character storage, + processing opn.)
Still want access to files even if computer is down.
- also - all info which goes into file must be recorded on tape - to be reconstructed in case of trouble —

Stretch properties:

- facilities for multiprog.
- I/O control units.

etc. - editing done internally

IDP — Feed in from many points — may come in faster than computer can handle it. written on tape by processor.

Tape is fed into computer later.

---

tape processor would have to have additional stand-by — in case of down.

-- (Stretch couldn't accept if down) --

— auditing — tape serves as record of transactions put into file — imp

"any old machine" which can read tape or write tape can work

in the system — manual switching of tapes in 2K.

Recorder — may not require stored prog as such.

— input comes from remote point (telephone line)

- cards
- plugs
- manual input

→ phone line → incremental type machine - would be ideal -

---

question of error correction on 7000-X:

7050 — why no checking of arithmetic itself — only check memory

How many errors can IDP stand?

— if check mechanical devices & have solid state.

then ~~cap~~ equipment would be 100 or 1/100 times more reliable than data being entered — (human errors), so can live with this reliability — but can't stand down periods.

IDP methods are more expensive than present methods - but IDP does jobs we can't do now. These must be economic for customer overall. such as decreased stocks + increased efficiency - customer goodwill - etc.

IDP — planning, recording, controlling — (at present one record only) "Total time" is the essence.

SABRE = an IDP — only full-scale project at the moment - commercial

SAGE - an IDP - military

Bank program - some study of machine use in savings bank.

---

2 Computers eg. 709's on same files. Amer. Airlines ?
how do they change machine ?

IDP: Computer: essentially batch processing
perhaps do a search for priority before doing each job from tape. — tape records chronologically only.

- purely tape fed - present records from disk + tape
output also tape
- interrogation to file only - depends on file computer - but may have to go thru computer to check all of conditions eg drawing a tool kit, bonus salary.

time: Transactions in — several minutes are OK.
requests - answer in seconds. (not milliseconds)

→ Tape: writing head → reading head (tape servo is buffer + log)
1 tape to each telephone line — may have several units on line.

many remote inputs are typical

250, were specified for Boeing Wichita ←

(cut to 350 later) proposal is being made, deliver summer '59

40 phone lines [at Sudbury Bough SE]

presently 5 characters per sec is proposed, best we can do now.

— can't justify computer for IDP only — want to be able to use any computer —

Internal characteristics of computer — same as any commercial application ←

— Boeing SE project = special small serial card reader — ignores Alaska etc.

question of mkt above 705 ?

IDP has not been proved yet.

Special applications first

— smaller systems will be put together & grow.

— in several years will require larger machines — can't justify now, but know on faith that they will grow.

— market exist there now (except for special cases) — it will develop with proof of systems in above cases. — an economic proof.